CSL605 Cloud Computing Lab **Sem VI**
Name: Hatim Mandviwala TE COMPS-B Roll no.:10

EXPERIMENT 1

Title	Study the importance of cloud computing and various types of deployment and service models.
Pre requisite	Computer Networks
Mapping with CO	CSL605.1
Objective	Understand deployment models, service models, and advantages of cloud computing.
Outcome	To discuss basics of cloud computing with the help of deployment and service models
Instructions	1. Soft copy submission for Activity 1 2. Hard copy submission for Activity 2 (Hand written)
Deliverables	**Activity 1:** **1. To understand the origin of cloud computing** **ANS:** Cloud computing originated from the evolution of **mainframe computing, client-server architecture, distributed computing, virtualization, and the internet**. In the early days, mainframe computers were shared by multiple users using time-sharing systems. Later, client-server models allowed users to access centralized servers through networks. With the growth of the internet in the 1990s, organizations started delivering computing services over networks. Virtualization technology allowed multiple virtual machines to run on a single physical machine, improving resource utilization. The concept of providing computing resources as services became popular in the early 2000s. Companies like **Amazon, Google, and Microsoft** started offering storage, servers, and applications over the internet. Thus, cloud computing emerged as a model where computing resources are delivered **on demand** over the internet with scalability and cost efficiency. **2. NIST model, characteristics of cloud** **ANS:** According to **NIST (National Institute of Standards and Technology)**, cloud computing provides **on-demand network access to a shared pool of**
	configurable computing resources that can be rapidly provisioned with minimal management effort.

	Essential Characteristics of Cloud Computing: 1. **On-Demand Self-Service** Users can automatically provision computing resources without human interaction. *Example:* Creating a virtual server instantly. 2. **Broad Network Access** Services are accessible over the network using standard devices like laptops and mobile phones. 3. **Resource Pooling** Cloud provider resources are shared among multiple users using a multi-tenant model. 4. **Rapid Elasticity** Resources can be quickly scaled up or down based on demand. 5. **Measured Service** Resource usage is monitored and billed on a pay-as-you-use basis. **3. Different deployment models** **ANS:** Deployment models define **how cloud infrastructure is deployed and who can access it**. **Public Cloud** • Owned and managed by third-party providers • Accessible to the general public • Cost-effective *Example:* AWS, Google Cloud, Microsoft Azure **Private Cloud** • Used by a single organization • Offers high security and control *Example:* Cloud used by banks or government organizations **Hybrid Cloud** • Combination of public and private clouds • Sensitive data remains private while other services use public cloud *Example:* Private database + public application hosting
	Community Cloud • Shared by organizations with similar requirements • Improves collaboration and cost sharing *Example:* Cloud shared by educational institutions or government departments **4. Service models**

	ANS: Service models define **what level of service is provided by the cloud vendor**. **Infrastructure as a Service (IaaS)** • Provides virtualized hardware resources like servers and storage • User manages OS and applications *Example:* Amazon EC2 **Platform as a Service (PaaS)** • Provides a platform to develop and deploy applications • User focuses on coding, not infrastructure *Example:* Google App Engine **Software as a Service (SaaS)** • Provides ready-to-use software over the internet • No installation or maintenance required *Example:* Gmail, Microsoft Office 365 **5. Cloud cube model** **ANS:** The **Cloud Cube Model**, proposed by the **Jericho Forum**, helps organizations analyze cloud adoption based on **security, ownership, and management responsibility**. It includes four dimensions: 1. **Internal vs External –** Cloud inside or outside the organization 2. **Proprietary vs Open –** Vendor-specific or open technologies 3. **Perimeterised vs De-perimeterised –** Traditional or boundary-less security 4. **Insourced vs Outsourced –** Managed internally or by a third party *Example:* A private internal cloud is **Internal + Proprietary + Perimeterised + Insourced**.

Activity 2:

Find one real-time problem or requirement and provide its non-cloud based solution and cloud-based solution.

Scenario: Smart Traffic Management System

🧾 **Problem Statement**

A city needs an intelligent system to monitor traffic flow, reduce congestion, and manage traffic signals dynamically using real-time data from cameras and sensors. During peak hours, the system must process massive data quickly to adjust signals and avoid traffic jams.

⚠️ **Problems Faced:**

- **Massive data generated from cameras and sensors**
- **Delays in processing real-time traffic data**
- **Traffic congestion due to slow decision-making**
- **High infrastructure cost for installing and maintaining local servers**
- **Difficulty in scaling the system as the city grows**

🖥️ **Non-Cloud-Based Solution**

The traffic management system is deployed on local government servers installed in a central control room.

⚙️ **How It Works**

- **Traffic cameras send data to a central local server**

- **Data is processed in one location**
- **Traffic signals are controlled manually or semi-automatically**
- **Storage is limited to local hardware**
- **System capacity depends on installed servers**

✅ Advantages

- **Full control over sensitive city data**
- **Works even with limited internet connectivity**
- **No dependency on external cloud providers**

❌ Disadvantages

- **Slow processing of real-time data**
- **Cannot scale easily as the city expands**
- **High maintenance and hardware costs**
- **System failure can disrupt entire traffic network**
- **Inefficient traffic signal optimization**

☁️ Cloud-Based Solution

The system is implemented using cloud platforms like Amazon Web Services, Google Cloud, or Microsoft Azure.

☁️ Cloud Services Used

	• **Infrastructure as a Service (IaaS)** • **Platform as a Service (PaaS)** • **AI/ML services for traffic prediction**
	⚙️ **How It Works** • **Traffic cameras stream data to the cloud in real time** • **Cloud systems process large volumes of data instantly** • **AI models predict congestion and optimize traffic signals** • **Auto-scaling handles peak traffic data loads** • **Dashboards provide live monitoring for authorities**
	🚀 **Advantages** • **Real-time traffic optimization** • **Scalable for growing cities** • **Reduced congestion using AI predictions** • **Lower infrastructure maintenance** • **Centralized monitoring with high availability**
	⚠️ **Limitations** • **Requires reliable internet connectivity** • **Data privacy and surveillance concerns** • **Continuous cloud costs**

Conclusion	**In today's digital era, systems like Learning Management Systems, food delivery platforms, and smart traffic management require high scalability, reliability, and real-time performance. Traditional non-cloud-based solutions, while offering full control and independence, struggle to handle large-scale data, sudden traffic spikes, and growing user demands.** **Cloud computing provides a more efficient and flexible alternative by offering on-demand resources, auto-scaling, and global accessibility. Platforms such as Amazon Web Services, Google Cloud, and Microsoft Azure enable organizations to store vast amounts of data, deliver content quickly, and maintain system stability even during peak usage.**
References	Amazon Web Services (AWS). *Cloud Computing Services*. Google Cloud. *Cloud Computing Services*. Microsoft Azure. *Cloud Computing Platform*. GeeksforGeeks. *Cloud Computing Tutorial*.